

HOSPITAL EMERGENCY ROOM
PATIENT ANALYTICS & DASHBOARD

Hospital Emergency Room Analytics Report

A Patient Flow, Wait Time & Satisfaction Study

Microsoft Excel Dashboard Project | 464 Patient Records | 2023-2024

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1. Overview

This report documents the design, build, and analytical findings of the Hospital Emergency Room Dashboard — a Microsoft Excel-based project developed to analyse patient flow, wait times, admission outcomes, and satisfaction levels within a hospital emergency department. The dashboard consolidates 464 patient visit records spanning 2023-2024 into a single analytical view for quick, evidence-based decision-making.

464 Total Patients	34.96 Avg. Wait Time (mins)	5.10 / 10 Avg. Satisfaction Score	231 Admitted Patients
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2. Introduction

Emergency room performance directly affects patient outcomes and satisfaction, making it essential for hospital administrators to track wait times, admission patterns, and department referrals closely. This dashboard brings together patient-level data — age, gender, wait time, satisfaction score, admission status, and department referral — into charts, KPI cards, and pivot summaries that make trends easy to spot and act on.

3. Objective

- To monitor patient flow through the emergency room across demographics and time.
- To analyse wait times and satisfaction scores in relation to admission outcomes and department referrals.
- To help hospital management identify bottlenecks, at-risk patient groups, and opportunities to improve patient experience.

4. Dataset Description

- **Source:** Hospital emergency room visit records (simulated for portfolio purposes)
- **Volume:** 464 individual patient records
- **Time Period:** January 2023 - December 2024
- **Coverage:** Patient demographics, wait times, satisfaction scores, admission status, and department referrals

5. Data Fields

- Patient ID, Date, Month, Year
- Age & Age Group
- Gender
- Patient Wait Time (minutes)
- Satisfaction Score (out of 10)
- Admission Status (Admitted / Not Admitted)
- Department Referral

- Attended On Time (On-Time / Delayed)

6. Dashboard Features & Visuals

- **KPI Cards:** At-a-glance totals for patient count, average wait time, average satisfaction score, and admission count.
- **Pivot Tables:** Breakdowns by gender, age group, admission status, department referral, and timeliness.
- **Bar & Pie Charts:** Visual comparison of patient volumes across demographics and outcomes.
- **Clean UI:** A single-page dashboard layout combining KPIs, charts, and tables for quick review.
- **Dashboard Tool:** Microsoft Excel (PivotTables, PivotCharts, xlswriter-rebuilt visuals)

7. Patient Demographics

The patient population is almost evenly split by gender (227 female, 237 male) and fairly evenly distributed across age groups, with the 60-69 bracket the largest single group (67 patients) and the 20-29 bracket the smallest (48 patients).

Gender Distribution

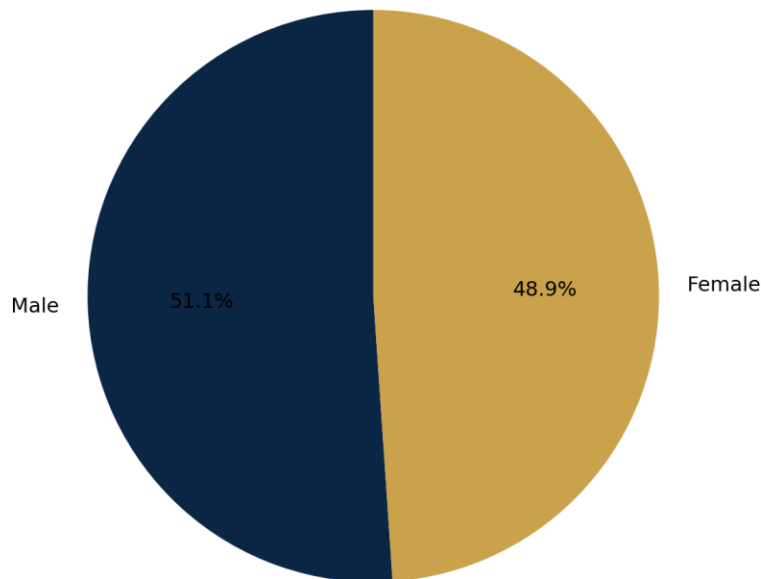


Figure 1: Gender Distribution

Patient Count by Age Group

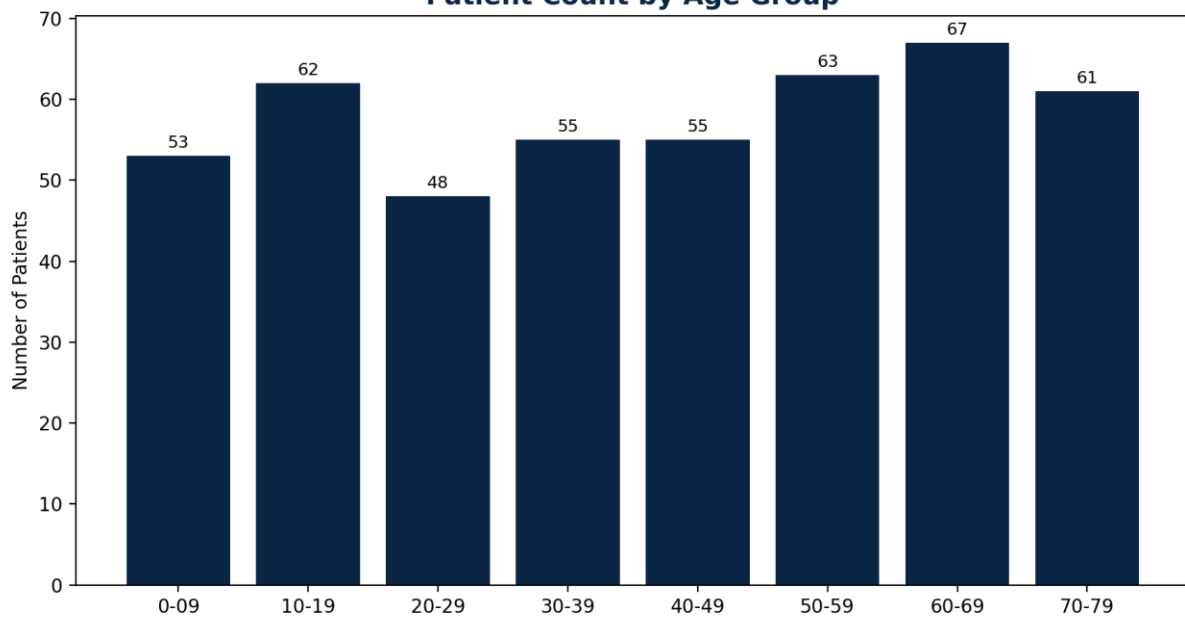


Figure 2: Patient Count by Age Group

8. Admission & Timeliness Analysis

Admission outcomes are almost evenly balanced, with 231 patients admitted and 233 not admitted. Timeliness shows more variance: 279 patients (60%) were attended on time, while 185 (40%) experienced delays — a meaningful share worth investigating further.

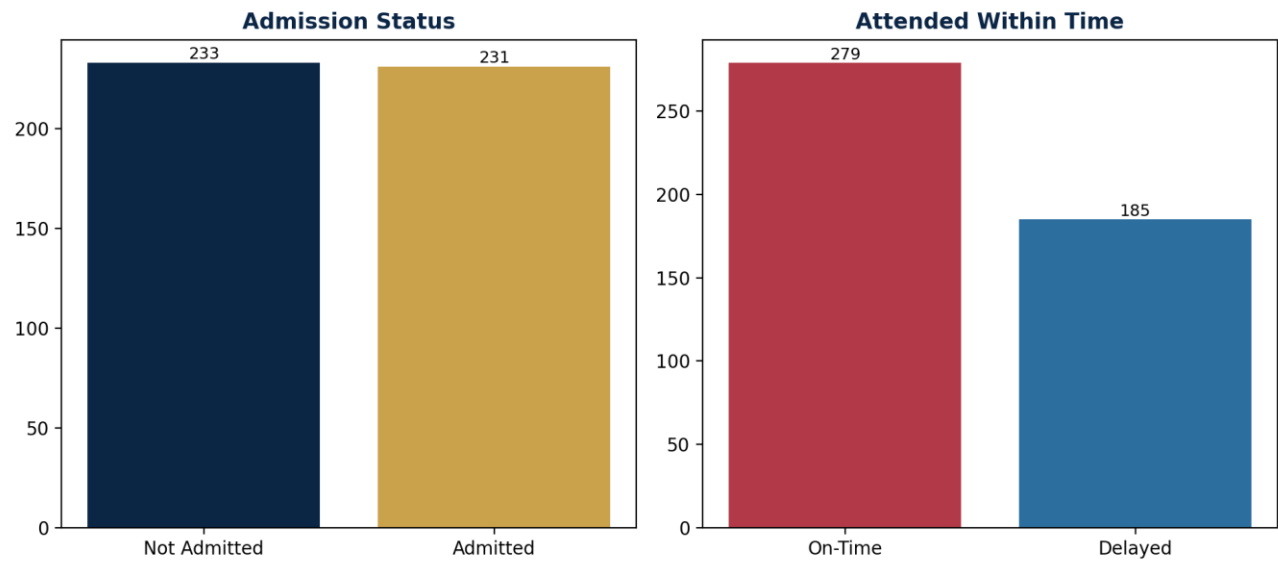


Figure 3: Admission Status & Timeliness of Attendance

Summary Table

Metric	Category	Count	% of Total
Admission Status	Admitted	231	49.8%
Admission Status	Not Admitted	233	50.2%
Attended On Time	On-Time	279	60.1%
Attended On Time	Delayed	185	39.9%

9. Department Referral Analysis

258 of 464 patients (55.6%) required no further department referral and were resolved within the emergency room itself. Among those referred onward, General Practice received the largest share by a wide margin, followed by Orthopedics.

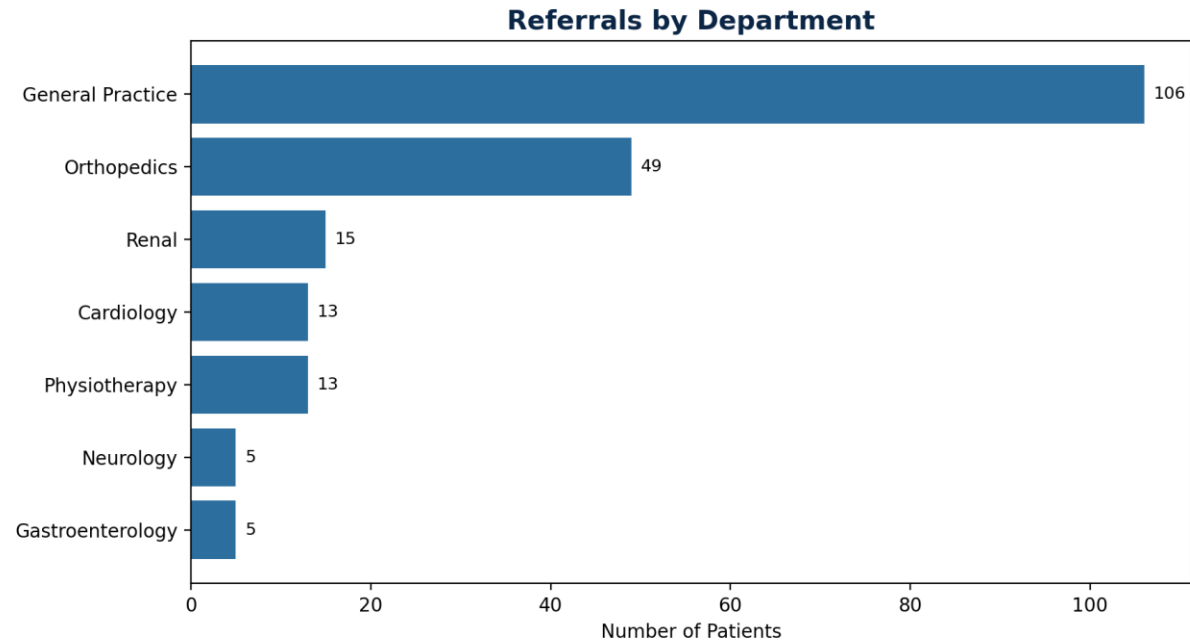


Figure 4: Referrals by Department (excluding no-referral cases)

Department	Referral Count	Avg. Wait Time (mins)	Avg. Satisfaction
General Practice	106	34.6	4.8
Orthopedics	49	34.1	5.3
Renal	15	35.0	5.6
Physiotherapy	13	34.5	6.2
Cardiology	13	33.5	4.7
Gastroenterology	5	41.2	5.7
Neurology	5	37.9	4.5

10. Wait Time & Satisfaction Analysis

Average wait time across all patients is 34.96 minutes (range: 5.2 to 67.9 minutes), and average satisfaction sits at 5.10 out of 10. Wait time and satisfaction show only a weak relationship with admission outcome or timeliness — satisfaction scores are nearly identical for admitted vs. not-admitted patients (5.04 vs. 5.16) and for on-time vs. delayed patients (5.08 vs. 5.14), suggesting satisfaction is driven more by factors like department and individual experience than by wait time or outcome alone.

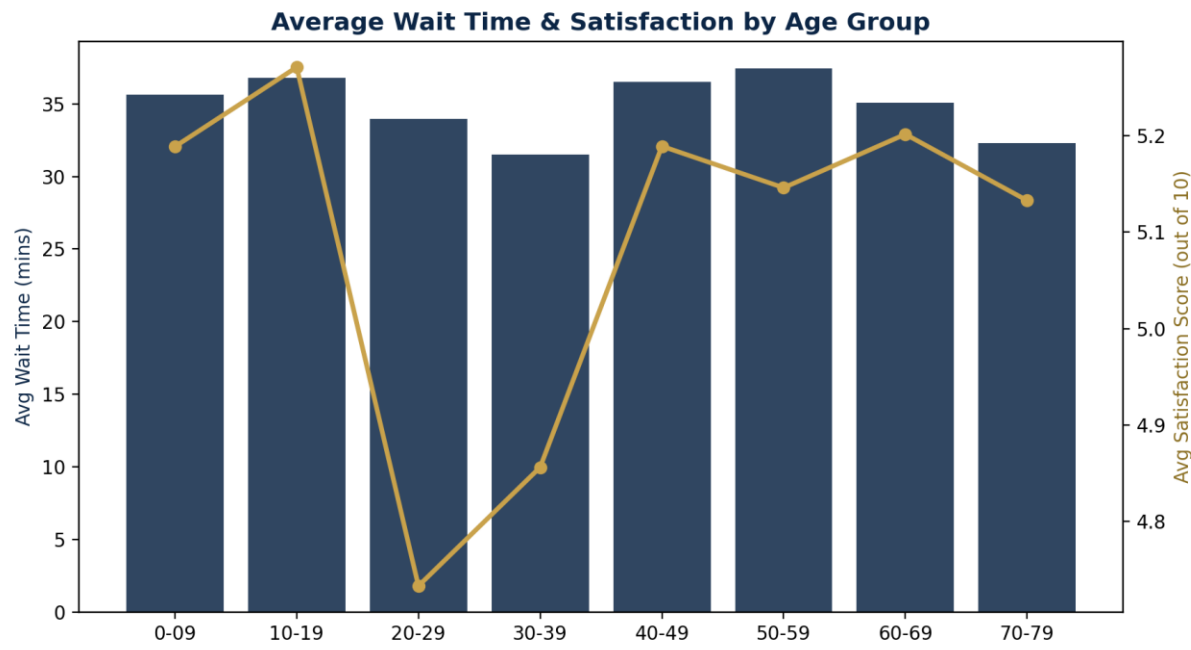


Figure 5: Average Wait Time & Satisfaction by Age Group

11. Key Insights

- Patient demographics are well balanced across gender and age group, with no single segment dominating ER volume.
- Nearly 40% of patients were attended with a delay, representing the clearest operational improvement opportunity in the dataset.
- General Practice accounts for the majority of onward referrals, suggesting it may be a bottleneck department worth resourcing further.
- Satisfaction scores remain fairly stable across admission outcomes and timeliness, indicating that patient experience is shaped by more than wait time alone.
- Physiotherapy and Renal referrals show the highest average satisfaction scores, while Neurology and Cardiology show the lowest — worth a closer qualitative look.

12. Conclusion

The Hospital Emergency Room Dashboard provides a clear, consolidated view of patient flow, admission outcomes, department referrals, and satisfaction across 464 emergency visits. It highlights where delays are concentrated, which departments carry the heaviest referral load, and where patient satisfaction is strongest and weakest. As a portfolio project, it demonstrates practical skills in pivot-table based analysis, KPI design, and dashboard storytelling in Microsoft Excel — directly relevant to healthcare and business analytics roles.

13. Future Enhancements

- Add month-over-month trend tracking for wait times and satisfaction.
- Introduce a root-cause breakdown for delayed attendance cases.
- Layer in staffing levels to correlate wait time with shift coverage.
- Build a predictive model to flag patients at risk of long waits or low satisfaction.